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Long-Term Educational Consequences of Vocational Training in Colombia ☑

Impacts on Young Trainees and Their Relatives

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ABSTRACT

Vocational training evaluations focus on trainees' earnings. This focus may understate these programs' benefits if training improves participants' and relatives' educational attainment. We use Colombian administrative data and a randomization design to examine the long-term employment and education impacts on trainees and their relatives. Eleven years after randomization, trainees increased higher education enrollments, and their relatives increased secondary school attainment. Training helped relax credit constraints for women, while improving field-specific knowledge for men. Including improved education impacts from training increases the program's estimated internal rate of return from 22.2 percent to 24.1 percent for females and from 10.2 percent to 25.5 percent for males.

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(continued on next page)

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I. Introduction

Vocational training programs typically aim to improve the employment prospects of individuals who face difficulties entering the labor force. For instance, vocational training often targets individuals who have dropped out of the formal education system before finishing secondary school and who typically have poorer employment prospects. The goal of many training programs is, thus, to provide participants with the skills needed to help them find gainful employment opportunities.

The literature evaluating the impacts of vocational training programs primarily focuses on the earnings of trainees. The evidence on whether these programs are attractive social investments—particularly in developing countries—is mixed. The net benefits of these programs may be understated, in part, because most evaluations of vocational training programs only examine the direct impact of these programs on earnings. This focus may understate these programs' overall benefits if training facilitates the acquisition of other forms of human capital for participants and their relatives, through the relaxation of credit or informational constraints.

The narrow focus on earnings outcomes in the training literature could stem from contextual and demographic factors, since training often targets older populations and displaced workers who may have limited prospects for further formal education (for example, Hirshleifer et al. 2016). In many settings, however, vocational training targets youth who have the opportunity to eventually attain further formal education (for example, Gelber, Isen, and Kessler 2016).¹ While previous studies have shown that skills beget skills in formal education (for example, Heckman 2000),² there has been

Amherst, George Washington University, the U.S. Bureau of the Census, the Development and America's Initiative workshops at Georgetown University, the 2016 NBER Labor Studies Spring Conference, the 2015 APPAM conference, the World Bank/IZA Employment Development Conference in Bonn, the LMK Impact Evaluation Workshop at the IDB, and the Human Development Capabilities Association Conference for helpful comments. The authors have nothing to disclose. This paper uses confidential SISBEN Census data, SABER 11 administrative data on graduation test scores from the Colombian Institute for the Promotion of Higher Education (ICFES), data from the System for the Prevention and Analysis of Dropouts in Higher Education Institutions (SPADIES), and Social Security administrative records from the Integral System of Social Protection Information (SISPRO). The SISBEN data can be obtained directly from the Colombian Department of National Planning. The SABER 11 test scores can be obtained from ICFES. The SPADIES data can be obtained from the Colombian Ministry of Education. The SISPRO data can be obtained from the Colombian Ministry of Health and Social Protection. The authors are willing to assist in contacting these various agencies to obtain access to the data (Adriana Kugler, ak659@georgetown.edu).

1. Gelber, Isen, and Kessler (2016) examine the impacts of a summer youth employment program for teenagers in New York City on college enrollment and find that the program does not affect subsequent earnings or college enrollment of participants. A related question is explored by Carrell and Kurlaender (2020), who assess the extent to which community college is a pathway to attainment of a four-year degree among U.S. students.

2. We examine the impact of a single intervention (that is, vocational training) on subsequent formal education. Other studies have examined complementarities of two different interventions. For example, Johnson and Jackson (2019) find evidence of dynamic complementarities in the United States in that the “benefits of Head Start spending were larger when followed by access to better-funded public K–12 schools, and the increases in K–12 spending were more efficacious for poor children who were exposed to higher levels of Head Start spending during their preschool years.” These findings suggest that investments in the skills of disadvantaged children that are followed by sustained educational investments over time can potentially break the cycle of

no systematic examination to date of whether vocational training begets additional formal educational investments among participants and whether there are educational spillovers on other family members.

This work makes three substantive contributions to the training literature. First, we document long-term impacts of training on the employment, earnings, and formal education outcomes of trainees. Second, we examine formal education spillovers on other family members. Third, we consider and empirically examine potential mechanisms through which training could have led to additional formal education investments of participants and relatives.

We study Youth in Action (YIA), a vocational training program for disadvantaged youth implemented at scale in the country of Colombia during the 2000s. This program provided participants with a three-month vocational classroom training module and a three-month private-sector apprenticeship. We leverage lotteries used to allocate over-subscribed training slots in the 2005 training cohort and combine various national administrative data sources to document impacts on participants and their relatives up to 11 years after randomization.³

Youth in Action vocational training increases access to formal tertiary education for trainees up to 11 years after program participation. Tertiary enrollment increases by 3.7 percentage points for men (from a 14.6 percent base) and by 3.2 percentage points for women (from an 11.5 percent base), though the gender difference is not statistically significant. Lottery winners and losers are equally likely to complete formal secondary school after program participation.

Some of the relatives of trainees also complete more formal schooling as a result of the program. Relatives of female lottery-winner trainees are 1.7 percentage points (from a 12.4 percent base) more likely to complete secondary school up to 11 years after training than relatives of female lottery losers.

These additional downstream formal education investments among trainees and their relatives could stem from informational externalities or from improved educational inputs and resources that participants gain access to during training, which allow them to relax household credit constraints. We conduct a number of formal empirical tests to try to shed light on three potential mechanisms. First, we examine whether vocational training helps relax credit constraints in households' ability to pay the direct costs of tertiary education; we use a version of the Cameron and Taber (2004) instrumental variables approach to test for educational credit constraints. Second, we examine whether trainees and their relatives learn about the value of general education through vocational training or alternatively obtain general skills through vocational training. Third, we examine whether trainees acquire specific skills or learn about the returns on those specific skills through their training. In particular, we use Markov transition

poverty. However, Malamud, Pop-Eleches, and Urquiola (2016) do not find evidence of complementarities between home and school environments in Romania.

3. Only a few randomized trials in developing countries have had long-term follow-ups more than ten years after random assignment. Maluccio et al. (2009) examine the impact of childhood nutrition program in Guatemala 25 years later. Kugler and Rojas (2018) follow CCT beneficiaries in Mexico up to 17 years later. Bettinger et al. (2018) explore the impact of secondary school vouchers for disadvantaged youth on labor market and other outcomes up to 17 years after the lottery. Barrera-Orsorio, Linden, and Saavedra (2019) examine long-term educational impacts of alternative CCT payment structures in Colombia. Baird et al. (2016) evaluate the deworming experiment in Kenya ten years after random assignment.

matrixes to shed light on the field specificity of training and formal education choices. The various empirical tests suggest that, for females, training helped relax credit constraints associated with direct tertiary education costs. For males, training appears to have improved field-specific knowledge and/or information about field-specific returns to tertiary education, as opposed to just general information or skills. For relatives, we mostly find evidence for females that may be consistent with spillovers arising from gender-specific informational externalities, which may be consistent with information sharing about gender identity and attitudes within the household (for example, Akerlof and Kranton 2000, 2010; Bertrand, Kamenica, and Pan 2015; Fernandez, Fogli, and Olivetti 2004; Fortin 2005, 2009).

In terms of labor market outcomes, among female and male applicants, winners are 6.5 percentage points and four percentage points, respectively, more likely to be formally employed than losers in the long run (from a 78.1 percent base for females and a 57.5 percent base for males). The short-term earnings of female and male lottery winners are higher than those of lottery losers. However, the long-term daily earnings of female and male lottery winners are not significantly higher than those of lottery losers. This is not because trainees do not have higher daily earnings in the long run, but rather because nontrainees' earnings catch up over time. In the short term, the program helps leapfrog trainees in the formal labor market. Over time, as nontrainees spend less time in formal education and presumably gain additional labor market experience, the earnings gap closes, but not completely, such that trainees are still better off in the long term.

To illustrate the welfare implications of the formal education impacts of vocational training, we conduct a cost-benefit analysis under three different scenarios: (i) only accounting for labor market impacts of participants accruing from vocational training participation, (ii) accounting for increased future earnings of participants due to increased completed tertiary education, and (iii) also accounting for increased future earnings of the relatives of participants due to increased secondary education. The internal rates of return (IRR) calculated on the basis of benefits from direct earnings effects on participants alone are 16.75 percent for the full sample, 22.1 percent for women, and 10.2 percent for men. Our IRR estimate of 16.75 percent, only accounting for direct labor market effects, is very similar to the 16 percent IRR reported by Attanasio et al. (2017). By contrast, accounting for the tertiary education impacts of vocational training among participants yields IRRs that are 23.5 percent for women and 20.5 percent for men. In addition, accounting for spillovers on other household members further increases the IRRs, to 24.1 percent for women and 25.5 percent for men. Thus, not accounting for these additional downstream educational investments of trainees and relatives substantially understates the social desirability of the Colombian vocational training program, particularly among men.

Our work is most closely related to other developing-country evaluations of training programs. Among the few rigorous training evaluations conducted in developing countries, Attanasio, Kugler, and Meghir (2011) find that in the short term, YIA increased the earnings of female participants by close to 20 percent and increased the formal employment participation of both females and males by up to seven percentage points after one year. A follow-up study finds that these early effects on labor market outcomes of the Colombian training program persist in the medium term (Attanasio et al. 2017). Attanasio, Kugler, and Meghir (2011) and Attanasio et al. (2017) find that the employment benefits of vocational training for women exceed the program costs,

even when assuming that skills depreciate over time. Field et al. (2019) evaluate the labor market impact of a randomized control trial of about 8,000 students in Mongolia between 2010 and 2012. Like Attanasio, Kugler, and Meghir (2011), they find that formal vocational secondary school enrollment significantly increased employment and earnings for women in the short term. However, evidence from the few other randomized controlled trials on vocational training in Argentina, the Dominican Republic, Kenya, and Turkey suggest modest short- and long-term impacts on the labor market outcomes of participants (Alzua, Cruces, and Lopez 2016; Card et al. 2011; Hicks et al. 2014; Hirshleifer et al. 2016; Ripani et al. 2018).⁴

In Section II, we describe the vocational training program in Colombia, prior evidence on its short-term employment effects, and related studies on the effects of vocational training for disadvantaged youth. In Section III, we discuss the administrative data sources we use for the analysis and how we linked them to original program application data. In Section IV, we present results on the long-term formal education impacts on participants and relatives, as well as effects on employment and earnings. In Section V, we present the results of tests to disentangle the potential channels through which these impacts work. We present welfare calculations in Section VI and conclude in Section VII.

II. Program Background and Prior Evidence

A. Program Background

We study the long-term direct and spillover effects of Youth in Action (YIA), a large-scale vocational training program introduced by the government of Colombia in the early 2000s. Between 2001 and 2005, more than 130,000 disadvantaged youth residing in Colombia's largest metropolitan areas received training through YIA. The YIA program was part of a social safety net strategy put in place by the Colombian government to support low-income families, who were severely affected by Colombia's 1999 recession. Targeting rules for each program ensured that participation in these three safety net programs did not overlap.⁵

4. Alzua, Cruces, and Lopez (2016) evaluate the impacts of a youth training program in Argentina that includes life skills and on-the-job training up to four years after random assignment. The study finds stronger short-term labor market effects for men than for women, yet the effects dissipate four years after random assignment. Card et al. (2011) analyze a training program in the Dominican Republic and find small positive impacts on formal employment and earnings. Hicks et al. (2014) examine the impact of an informational and training voucher intervention in Kenya and find that vouchers increased training participation but did not increase participants' earnings. Hirshleifer et al. (2016) study the impact of a randomized training program in Turkey and find no positive effect on employment or earnings. That study does find a positive effect on employment quality one year after the program, but finds that the effect disappears three years later. Ripani et al. (2018) find small long-run employment and earnings impacts of the Dominican Republic training program.

5. The other two programs were Families in Action and Employment in Action. Families in Action provided conditional cash transfers to poor rural households that kept their children in school and took them to receive health center checkups. Employment in Action was a workfare program offering public employment to adults. While YIA targeted disadvantaged youth in urban areas, Families in Action targeted rural poor, displaced, and indigenous populations in towns with less than 100,000 inhabitants. Employment in Action mainly targeted poor adults in rural areas and smaller cities (63 percent of participants lived in cities with less than 100,000

Youth in Action targeted socioeconomically disadvantaged youth residing in Colombia's seven largest metropolitan areas—Barranquilla, Bogotá, Bucaramanga, Cali, Cartagena, Manizales, and Medellín.⁶ Applicants had to be between 18 and 25 years old and out of school or unemployed at the time of application to be eligible to receive services.

The YIA program included two unique features. First, YIA combined a three-month classroom-training module with a three-month apprenticeship in a formal private-sector job. The apprenticeship component is not a standard feature of many vocational training programs offered through schools. A second unique feature of the program was the pay-for-performance system, in which training institutions were only paid if an applicant finished the three months of training and was then placed in an apprenticeship.

In 2005, 114 legally registered, financially solvent training firms, selected in a competitive bidding process, offered 441 training courses (989 classroom sections in total).⁷ Trainees spent between seven and eight hours per day, five days per week, on classroom training. Classroom training combined job-specific skills with noncognitive life skills. Trainees received a \$2.20/day stipend to cover transportation and other expenses during classroom and apprenticeship modules. Female trainees with children under age seven received a higher stipend of \$3.00/day.

Upon completion of the classroom-training module, training firms had to place trainees in formal-sector apprenticeships. Payments to training firms were conditional on verified apprenticeship placement. This pay-for-performance element of the program created incentives for training firms to offer training that was relevant to employers' needs. In 2005, 1,009 formal-sector companies offered apprenticeships to YIA participants. On average, trainees spent 5.19 hours per day participating in manufacturing, retail, trade, and service-sector apprenticeships. Throughout the classroom and apprenticeship modules, trainees received self-esteem and self-advocacy mentorships through an enrichment program called Life Project (in Spanish, Proyecto de Vida).

In 2005, the government used lotteries to determine eligibility for YIA participation. For each course offered, training firms were asked to pre-select up to 50 percent more eligible applicants than the course's enrollment capacity (for example, a course with a capacity for 30 individuals would pre-select a list of up to 45 individuals). Two-thirds of applicants from these course-specific pre-selected lists were randomly assigned to receive training.⁸ In 2005, 33,284 eligible youth applied for training through YIA, 19,495

inhabitants). Moreover, Employment in Action operated between 2000 and 2004, before the YIA program participation of the 2005 cohort, which we study here.

6. The exact targeting criteria included being in the socioeconomic strata 1 or 2 of SISBEN, which is a national classification system used by the Colombian government to create means tests to target national welfare programs. This SISBEN targeting corresponds roughly to the lowest two deciles of the income distribution.

7. The courses include a wide variety of occupations, including inventory and warehouse assistant, electrician, archival assistant, pharmacy assistant, clinical lab assistant, auto mechanic assistant, welding assistant, data entry assistant, upholstery, preschool teacher assistant, beautician, call center assistant, agricultural machinery mechanic, seamstress, organic waste processor, flower cultivation, metal fabrication, meat processor, shoe repair services, florist, wooden machine operator, bank teller, and physical rehabilitation.

8. Attanasio, Kugler, and Meghir (2011) carried out the random assignment with the help of a firm in Colombia, which conducts surveys and statistical analysis. Using a random number generator, applicants were assigned to a number between 1 and the maximum number of applicants in the randomization pool in each

of whom were women. Among eligible applicants, 28,021 (84 percent) randomly received an offer to participate; women and men had similar odds of being randomized into the program.

There was close to full compliance with the randomization protocol in the 2005 training cohort. Only 0.18 percent of youth who were offered training turned it down, and only 1.29 percent of those who were not offered training ended up receiving training through YIA. Ninety-seven percent of participants completed the classroom-training module; 92 percent also completed the job apprenticeship. The YIA applicants randomly assigned to the control group did not receive other forms of publicly funded training, as no other public training program for disadvantaged youth existed in Colombia at the time.⁹ In our analyses, we use the original random assignment to estimate intent-to-treat (ITT) effects, comparing those randomly assigned to a spot in a course and those randomly assigned to the waiting list or control group in the same course.

B. Prior Evidence on the Effects of YIA

Attanasio, Kugler, and Meghir (2011) show that, one year after the lottery, female training lottery winners are seven percentage points more likely to be in formal-sector employment and earn 20 percent more than lottery losers. Among males, YIA increased formal employment by six percentage points but had no effect on earnings. Using the same randomization cohort as we do, Attanasio et al. (2017) document persistent earnings gains and increased likelihood of formal-sector employment in the medium term among program participants. On the basis of these impacts and program cost data, Attanasio et al. (2017) estimate the program's internal rate of return of 16 percent—to preview, this is close to our calculation of 16.75 percent, which only accounts for direct earnings effects, as Attanasio et al. (2017) do. We estimate an IRR of 22.1 percent among women and 10.2 percent among men without accounting for educational spillovers from vocational training; 54.8 percent of our sample is female, which yields an IRR estimate of 16.75 percent for the full sample, as reported above. This calculation, however, likely understates the social desirability of YIA because it does not account for either the potential impact of training on educational attainment of trainees or for spillovers from trainees to relatives, both of which we account for in this paper.

course. Those assigned numbers between 1 and the number of spots assigned through randomization in each course were offered a place, and those with a number above would be placed on a waiting list. Since Attanasio, Kugler, and Meghir (2011) randomized applicants in all courses, they could monitor whether someone tried to sign up for another course or eventually was accepted into the treatment group for a course. Attanasio, Kugler, and Meghir (2011) also monitored changes in participation status month-by-month among those who entered the randomization pool.

9. While the National Learning Service (Servicio Nacional de Aprendizaje, SENA) is a public entity under the Colombian Ministry of Labor offers vocational training, the vast majority of its course offerings provide training to currently employed workers whose employers contribute payroll taxes that finance the SENA. In fact, to register in SENA courses, one has to have a registration number provided by an employer that is included in a list of qualifying employers. SENA also provides some courses to disadvantaged populations (for example, incarcerated populations) and those in the informal sector, but these offerings are much more limited in number and by location.

III. Data Sources

A. Randomization Data

The sample for our analyses consists of a random sample of applicants from the 2005 YIA training cohort originally collected by Attanasio, Kugler, and Meghir (2011). Among all 2005 training applicants in the randomization pool, Attanasio, Kugler, and Meghir (2011) collected baseline information for a sample stratified by initial treatment offer, city, and gender. This baseline sample includes 2,041 applicants randomly assigned to the treatment group and 1,913 assigned to the control group.¹⁰ Attanasio, Kugler, and Meghir (2011) collected baseline data in January 2005, either before the beginning of the training program or during the first week of classes, to minimize any influence of participation in the program on interviewees' responses. This randomization data contain complete information on baseline characteristics for all sampled applicants.

As in Attanasio, Kugler, and Meghir (2011) and Attanasio et al. (2017), baseline characteristics of training applicants in the 2005 cohort are fairly balanced across randomization groups. Table 1 reports coefficients of regressions of the variable on a treatment indicator and training institution fixed effects to account for the randomization structure. As a result of the sample stratification by gender, 54.8 percent of lottery losers are female (even though two-thirds of applicants in 2005 are female). Lottery losers are, on average, 21 years old at the time of application, and 20 percent of them are married. Of lottery losers, about 24 percent have finished secondary schooling, and a little more than 5 percent have enrolled in tertiary schooling prior to the program, numbers that are not significantly different from those of participants in the program. At the time of application, about 20 percent of lottery losers report being employed, and 8.5 percent report being employed in the formal sector. These employment rates are not significantly different from those of lottery winners. Average tenure before training is a little more than three months. Lottery losers work, on average, 12 days per month (including zeros for those out of work) and 25 hours per week (including zeros for those out of work).

The only statistically significant difference in baseline characteristics among female lottery winners and losers is job tenure. Women lottery winners have, on average, slightly longer tenure in their previous job. For men, there are differences in marital status between the treatment and control groups, and age and hours worked are marginally significantly different. Male training lottery winners are, on average, less likely to be married. Importantly, secondary school completion, tertiary school enrollment, employment, and earnings are all insignificant. The fact that these placebo tests are insignificant makes any differences in these outcomes after training more credible. For women, the characteristics of the treatment and control groups are only marginally jointly statistically different from each other at the 10 percent level; for men, they are jointly significant at the 5 percent level. The *F*-test of joint significance reported at the bottom of Panel A in Table 1 comes from a regression of the treatment indicator on all

10. There were originally 4,351 individuals in the treatment and control groups of the baseline sample, but as in Attanasio, Kugler, and Meghir (2011), we drop 9.1 percent of the individuals in the sample, who were randomly assigned to the treatment and control groups after the January 18, 2005 date when all random assignments were supposed to occur. This does not introduce any bias, as we use the original random assignment for the 3,954 observations used for all our analysis.

Table 1
Baseline Characteristics of Applicants and Randomization Balance, Various Data Sources

	All Applicants		Males		Females		Male-Female Difference <i>p</i> -Value (7)
	Control Mean (SD) (1)	Participant Effect (SD) (2)	Control Mean (SD) (3)	Participant Effect (SD) (4)	Control Mean (SD) (5)	Participant Effect (SD) (6)	
Panel A: Baseline Data							
Female	0.548	-0.017 (0.016)					
Age	21.2	-0.158** (0.065)	21	-0.165* (0.094)	21.4	-0.141 (0.089)	0.849
Married	0.202	-0.016 (0.013)	0.128	-0.037** (0.015)	0.262	0.007 (0.019)	0.066
Monthly wage	53.4	3.13 (2.65)	63.6	3.42 (4.27)	45	2.29 (3.27)	0.834
Monthly earnings ^a	12.5	-0.426 (1.36)	18.5	-3.31 (2.44)	7.43	1.77 (1.42)	0.072
Formally employed	0.085	0.012 (0.009)	0.11	0.009 (0.015)	0.064	0.014 (0.011)	0.819
Has work contract	0.085	-0.001 (0.009)	0.103	-0.005 (0.014)	0.07	0.001 (0.011)	0.736
Job tenure (months)	3.24	0.765*** (0.275)	3.77	0.535 (0.381)	2.81	0.938** (0.395)	0.462

(continued)

Table 1 (continued)

	All Applicants		Males		Females		Male-Female Difference <i>p</i> -Value (7)
	Control Mean (SD) (1)	Participant Effect (SD) (2)	Control Mean (SD) (3)	Participant Effect (SD) (4)	Control Mean (SD) (5)	Participant Effect (SD) (6)	
Days/month worked	12.1	0.693* (0.404)	13.6	0.759 (0.593)	10.8	0.549 (0.547)	0.794
Hours/week worked	24.9	1.85** (0.91)	28.2	2.24* (1.35)	22.2	1.31 (1.22)	0.610
Completed secondary school before training	0.241	0.002 (0.014)	0.25	0.013 (0.02)	0.234	-0.008 (0.018)	0.458
Initial tertiary enrollment before training	0.053	-0.004 (0.007)	0.061	-0.008 (0.011)	0.047	-0.002 (0.009)	0.651
Applicant is working	0.201	-0.017 (0.013)	0.24	-0.02 (0.02)	0.169	-0.016 (0.016)	0.868
<i>F</i> -test of joint significance of baseline characteristics	2.220		2.000		1.610		
<i>p</i> -value	0.007		0.021		0.081		
Panel B: Match Information							
Applicant has valid national ID number	0.961	0.007 (0.006)	0.918	0.016 (0.012)	0.997	0.002 (0.002)	0.251
Matched to 2005 SISBEN census of the poor	0.881	-0.006 (0.01)	0.862	0.01 (0.016)	0.897	-0.019 (0.014)	0.170

(continued)

Table 1 (continued)

	All Applicants		Males		Females		Male-Female Difference <i>p</i> -Value (7)
	Control Mean (SD) (1)	Participant Effect (SD) (2)	Control Mean (SD) (3)	Participant Effect (SD) (4)	Control Mean (SD) (5)	Participant Effect (SD) (6)	
Matched to SABER 11 secondary graduation data	0.302	0.014 (0.015)	0.298	0.029 (0.022)	0.304	0.002 (0.02)	0.362
Matched to SPADIES tertiary enrollment data	0.182	0.029** (0.013)	0.207	0.024 (0.019)	0.162	0.032* (0.017)	0.768
Matched to SISPRO Social Security records	0.71	0.063*** (0.014)	0.817	0.024 (0.018)	0.622	0.091*** (0.02)	0.013
<i>F</i> -test of joint significance of baseline characteristics	4.790		0.941		4.550		
<i>p</i> -value	0.000		0.453		0.000		
Observations	3,954		1,822		2,132		

Notes: The table reports control group means and differences between participants and controls among all applicants and separately by gender. Variables used are from the baseline survey collected by Attanasio, Kugler, and Meguir (2011) from a random sample of applicants from the 2005 training cohort, Colombia's secondary school graduation exam (SABER 11) database for secondary graduation variables, and Colombia's Education Ministry's System for Prevention and Analysis of Dropouts in Higher Education (SPADIES) database for tertiary education variables, and Colombia's 2005 SISBEN census of the poor. Participant effect estimates are from separate OLS regressions for each characteristic that also include training institution fixed effects. Wages and earnings are expressed in 2013 USD (1 USD = 1869.1 Colombian pesos). Wages, earnings, formal, contract, days, and hours are zero if the applicant is out of work. Job tenure is expressed in months. Days worked are expressed per month, and hours worked are expressed per week. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

a. From self-employment.

the characteristics included in Panel A.¹¹ While the treatment and control groups are largely similar, in all the analyses that follow, our results include the full set of characteristics in Panel A. We also conduct robustness checks that include and exclude those baseline controls that show significant differences in Panel A of Table 1.

We use applicants' full names, date of birth, and adult national identification numbers to match the original randomization data to four government administrative data sets.¹² Matching to administrative data enables us to obtain identifying information on relatives and to track those who registered in the program through administrative education and Social Security records. This means that we can follow the educational and labor market outcomes of participants (and their relatives) many years after random assignment. In the 2005 applicant list, 100 percent of applicants report their full names, and more than 96 percent report a valid adult identification number.¹³ Most importantly, the first row in Panel B of Table 1 shows that there is no difference between lottery winners and losers in the probability of having a valid adult identification number. This implies that there is no prior selection in the ability to match treatment and control individuals to education and labor market data.

B. Baseline Identifying Information for Relatives

We obtain baseline information on applicants' relatives by matching the original randomization data to Colombia's 2005 census of the poor (known as 2005 SISBEN). The Colombian government uses the SISBEN census data to determine eligibility for all government subsidies. The 2005 SISBEN census covered about 60 percent of all households in Colombia at the time. From the SISBEN census we obtain identifying information for all relatives residing in the same household as YIA program applicants in 2005, which corresponds to the baseline year for the 2005 randomization cohort.

Relatives' identifiers include full names, dates of birth, national identification numbers, and relationship to the applicant. The second row in Panel B of Table 1 shows that we match 87.5 percent of lottery winners' relatives and 88.1 percent of lottery losers' relatives to the 2005 SISBEN census. The match rates between lottery winners and losers are statistically indistinguishable, indicating no differential selection in the relatives we are able to follow from the two randomization groups. While we identify relatives living in the household of the beneficiary prior to participation in the program, we can follow these relatives during and after participation in the program even after they have moved out of the households.¹⁴

11. Note that Panel A includes measures of education from administrative records, including completed secondary and enrolled in tertiary education prior to training, as well as administrative data on whether the applicant is working.

12. The national identification number or "cédula de ciudadanía" is similar to the Social Security Number (SSN) in the United States. This identification number allows individuals to register in all government programs and vote in Colombia.

13. Valid identification numbers have at least eight digits.

14. Relatives are a selected sample of those living in the household before the program starts, but they are not restricted to being in the household during and after participation in the program. This means that if there is selection of relatives, it is based on having lived in the household prior to the program. This may restrict the sample to children and young adults, who are more likely to still be going to school. However, it may also include older family members and less economically independent relatives, who may be less likely to go to school. Thus, there is no clear way to predict which way the selection will bias the effects of the program for this sample of relatives.

Table 2 shows the characteristics of the relatives identified in the 2005 SISBEN census. About 51 percent of relatives in the control group are female. There are seven people, on average, in the households of those in the control group, and about 17 percent are children of the beneficiary, 31 percent are siblings, 3 percent are spouses, and 48 percent have another relationship to the beneficiary. Of those in the control group, 33 percent are younger than 15, and 79 percent are younger than 45 years of age. Only 6 and 5 percent of male and female relatives of those in the control group completed high school, and only 2.5 and 2.7 percent of male and female relatives ever enrolled in higher education. The characteristics of relatives in the treatment and control group are mostly very similar. Household size is smaller for female participants, and the fraction of female relatives is smaller, though the difference is only marginally significant. For men, household size is smaller, and they are less likely to have children of their own in the household. While both female and male relatives are similar prior to training, in terms of secondary school completion, and female relatives are similar in terms of tertiary school enrollment, male relatives are more likely to be enrolled in higher education prior to the training program. These preexisting differences in relatives' education prior to training provide placebo tests. Thus, for female relatives, the placebos make more credible any results we find *ex post*. For male relatives, we have to be cautious of any treatment-control difference in tertiary schooling post-training, since the difference already existed pre-training.

C. Administrative Secondary and Tertiary Education Records

To track applicants' and relatives' educational outcomes, we use two national administrative data sets. First, we use Colombia's secondary school graduation exam (the SABER 11 exam) database. The SABER 11 exam is a standardized test, similar to the American SAT, that is offered by the government twice a year to students in their last year of high school.¹⁵ We use these data to determine whether applicants took the SABER 11 test, which we interpret as a proxy for secondary school graduation, since taking the test is a graduation requirement. We match data from the exam-taking cohort of 2000 through the exam-taking cohort of 2016, 11 years after applying to YIA.

Taking the test is, therefore, the relevant outcome of interest because we code secondary school graduation as one if the applicant appears in the SABER 11 data set and zero if not. Therefore, the secondary school completion variable is well defined for the entire sample, and the sample is not selected based on having outcome data. We match 30 percent of those in our original sample with SABER 11 data, and the differences between treatment and control groups are not significantly different from each other, as shown in the second row of Panel B in Table 1. Note that this match also represents the outcome of interest, since no match indicates that the person did not take the exam and did not graduate from secondary school. As explained above, since we are also able to match applicants (and relatives) to SABER 11 data before program participation, we can

15. The SABER 11 exam used to be called the ICFES exam, which took the acronym of the Instituto Colombiano de Fomento para la Educación Superior (Colombian Institute for the Promotion of Higher Education), which develops and offers the exam. The ICFES still administers the exam, ensures the quality of high schools, and provides information to universities on students' preparation in five subjects (math, reading and writing in Spanish, social science, natural sciences, and English).

Table 2
Baseline Characteristics of Relatives in Applicants' Households

	All Applicants		Males		Females		Male-Female Difference <i>p</i> -Value (7)
	Control Mean (SD) (1)	Participant Effect (SD) (2)	Control Mean (SD) (3)	Participant Effect (SD) (4)	Control Mean (SD) (5)	Participant Effect (SD) (6)	
Panel A: Household Beneficiaries Survey							
Female relative	0.509	-0.013 (0.008)	0.458	0.001 (0.012)	0.548	-0.021* (0.012)	0.178
Household size	7.05	-0.402*** (0.058)	6.9	-0.342*** (0.085)	7.16	-0.441*** (0.078)	0.395
Relative older than sample mean applicant age	0.519	0.002 (0.008)	0.557	0.002 (0.012)	0.489	0 (0.011)	0.869
Relative younger than 15 years old	0.335	-0.002 (0.008)	0.286	-0.004 (0.011)	0.372	0.003 (0.01)	0.640
Relative younger than 30 years old	0.629	0.005 (0.008)	0.591	-0.001 (0.012)	0.659	0.013 (0.01)	0.380
Relative younger than 45 years old	0.788	0.001 (0.007)	0.763	0.005 (0.01)	0.807	-0.001 (0.008)	0.614
Spouse/partner	0.033	0 (0.003)	0.013	0.001 (0.003)	0.048	0.001 (0.005)	0.956

(continued)

Table 2 (continued)

	All Applicants		Males		Females		Male-Female Difference <i>p</i> -Value (7)
	Control Mean (SD) (1)	Participant Effect (SD) (2)	Control Mean (SD) (3)	Participant Effect (SD) (4)	Control Mean (SD) (5)	Participant Effect (SD) (6)	
Son/daughter	0.171	-0.012** (0.006)	0.116	-0.019** (0.007)	0.213	-0.002 (0.009)	0.152
Sibling	0.313	0 (0.007)	0.364	-0.007 (0.011)	0.275	0.003 (0.01)	0.518
Other relationship	0.483	0.012 (0.008)	0.507	0.025** (0.012)	0.465	-0.001 (0.011)	0.108
Panel B: Secondary and Tertiary Administrative Data							
Completed secondary school before training (all relatives)	0.056	0.006 (0.004)	0.062	0.005 (0.006)	0.052	0.005 (0.005)	0.978
Completed secondary school before training (male relative)	0.061	0.006 (0.006)	0.069	0.003 (0.008)	0.054	0.008 (0.007)	0.624
Completed secondary school before training (female relative)	0.052	0.005 (0.005)	0.055	0.009 (0.008)	0.05	0.002 (0.006)	0.554

(continued)

Table 2 (continued)

	All Applicants		Males		Females		Male-Female Difference <i>p</i> -Value (7)
	Control Mean (SD) (1)	Participant Effect (SD) (2)	Control Mean (SD) (3)	Participant Effect (SD) (4)	Control Mean (SD) (5)	Participant Effect (SD) (6)	
Initial tertiary enrollment before training (all relatives)	0.026	0.008*** (0.003)	0.025	0.014*** (0.004)	0.026	0.003 (0.004)	0.042
Initial tertiary enrollment before training (male relative)	0.025	0.011*** (0.004)	0.026	0.019*** (0.006)	0.024	0.004 (0.005)	0.060
Initial tertiary enrollment before training (female relative)	0.027	0.006 (0.004)	0.025	0.01 (0.006)	0.028	0.003 (0.005)	0.380
Observations	7,902		3,215		4,687		

Notes: The table reports control group means and differences between participants and controls among all applicants and separately by gender of characteristics of applicant relatives residing in the same household at baseline. In Panel A data source is SISBEN 2005 census of the poor. In Panel B data sources are the SABER 11 database for secondary school graduation and the SPADIES database for tertiary education enrollment. Participant effect estimates are obtained from separate OLS regressions for each characteristic that also include training institution fixed effects. Wages and earnings are expressed in 2013 USD (1 USD = 1869.1 Colombian pesos). Wages, earnings, formal, contract, days, and hours are zero if the applicant is out of work. Job tenure is expressed in months. Days worked are expressed per month, and hours worked are expressed per week. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

conduct placebo tests examining secondary school completion outcomes before the implementation of YIA, when we should not observe any effects.

To track tertiary enrollment outcomes of applicants and relatives, we use the Colombia Education Ministry's Sistema de Prevención y Análisis de la Deserción en Instituciones de Educación Superior, or SPADIES (System for Prevention and Analysis of Dropouts in Higher Education Institutions). We refer to SPADIES data as the tertiary education database. The tertiary education database is an individual-level panel data set that tracks 95 percent of tertiary education freshmen up to their degree receipt (or dropout), beginning in 1998.¹⁶ We obtained data through 2016, 11 years after program application.

As with the secondary graduation data, appearing in the tertiary education database is the relevant outcome of interest. We code tertiary education enrollment as one if the applicant appears in the tertiary data set and zero if not. Thus, it is not surprising that the match rate with our original sample is 18.2 percent and is slightly higher for those in the treatment group, as shown in the fourth row of Panel B in Table 1. The tertiary enrollment variable is well defined for the entire sample, and the sample is not selected based on having outcome data. We interpret differences between lottery winners and losers in the probability of being matched to the tertiary education data post-randomization as a tertiary enrollment effect resulting from random assignment to YIA. As explained above, since we are also able to match applicants (and relatives) to the tertiary education data set for some years before program participation, we can conduct placebo tests examining tertiary enrollment outcomes before the implementation of YIA, when we should not observe any effects.

D. Social Security Records

To track formal employment and earnings outcomes, we match applicants to Colombia's Social Security records collected by the Ministry of Health and Social Protection, known as SISPRO (Sistema Integral de Información de la Protección Social).¹⁷ SISPRO is an individual-level monthly payroll tax panel data set. It contains information on whether individuals worked in the formal sector in a given month, the number of days of formal-sector employment, monthly earnings, and payroll taxes. We focus on SISPRO outcomes for 2008–2013, between three and eight years after randomization. We use data only starting in 2008, the year that SISPRO began to cover the universe of formal-sector workers. We focus on the following SISPRO variables: (i) appearing in the SISPRO database and (ii) average daily total earnings. Note that these outcomes are well defined for the entire sample because workers not matched to the SISPRO database in a given month are not in the formal sector that month, implying that formal earnings for that month are zero.¹⁸ The last row of Panel B in Table 1 shows that the match rate is 71

16. The tertiary education database is similar to the National Student Clearinghouse in the United States.

17. The SISPRO database only includes people who worked for employers that register their workers in the Social Security System or self-employed workers who register themselves.

18. The average daily earnings are constructed as follows. For each applicant (and relative), we first compute annual formal employment days as the sum of formal days per month (zeros included for months without formal employment) for each year 2008–2013. We follow the same procedure for annual formal earnings. For each applicant and relative, we then create an annual average of formal employment days and formal earnings for 2008–2013, our main variables of interest. Daily formal earnings are annual earnings divided by annual formal-sector employment days. Throughout, we express monetary values in 2013 U.S. dollars.

percent, which means that at some point during the 2008–2013 period, those in our randomized sample engaged in formal employment, our outcome of interest. Also, the last row shows that all lottery winners and female lottery winners are more likely to be matched to SISPRO, which reflects the positive impact of the program on formal employment.

One limitation of the SISPRO data is that we can only measure formal labor market outcomes, particularly employment and earnings. To estimate total earnings, we combine SISPRO data with nationally representative household survey data from Colombia's Gran Encuesta Integrada de Hogares (GEIH). For comparability with the YIA population, we restrict 2008 and 2010 GEIH data to individuals from the lowest socioeconomic strata, who would have been age-eligible at the time of YIA application and who reside in one of the seven metropolitan areas targeted by YIA. Separately for each of the 112 cells resulting from age-at-application (18–25, or eight age groups), gender (male, female), and metropolitan-program-area (seven metropolitan areas) combinations, we calculate annual informal earnings and the probability of informal employment. For each YIA applicant, we then compute their annual average total earnings as the weighted average of annual formal earnings multiplied by the likelihood of formal employment for that applicant's age–gender–metropolitan area–treatment status cell and annual informal earnings multiplied by the likelihood of informal employment in the corresponding cell. Note that this imputation approach assumes the same informal-sector earnings for both treated and control applicants. However, Attanasio, Kugler, and Meghir (2011) find that lottery winners, particularly females, earn higher informal earnings than lottery losers in the short term. Therefore, our imputed daily total earnings impact estimates represent a lower-bound treatment effect.

IV. Long-Term Training Impacts on Education and Labor Market Outcomes

In this section, we describe long-term training impacts on participants' and their relatives' secondary school completion, tertiary school enrollment, and employment and earnings. In Section IV.A, we present results on direct effects on secondary school completion and tertiary education enrollment among vocational training participants. In Section IV.B, we present results on indirect effects on relatives' secondary school completion and tertiary education enrollment. We then present the results of the impacts of YIA on employment and earnings of participants and their relatives in the long run in Section IV.C.

We compare all outcomes by original randomization status, so that all estimates are ITT effects. All specifications reported below in this and later sections control for the baseline characteristics in Panel A of Table 1 (observed for all applicants), including: gender, age, marital status, employment status, salary, profits, formal employment status, whether the person had a written contract, tenure, days worked, hours worked, whether the person completed secondary school, and whether the person enrolled in tertiary education. This is important, since there were a few differences between the treatment and control groups at baseline. We also control for training institution fixed effects to account for the randomization structure, implying that we are using random

variation in treatment assignment status into a course within a training institution. This is important, since individuals can self-select into training institutions, but not into treatment status.

A. Impacts on Participants' Secondary School Completion and Tertiary Enrollment

Twenty-five percent of male program applicants and 23.4 percent of female program applicants completed secondary school before training. There are no differences in secondary school completion rates among training lottery male and female winners and losers before training (Panel A of Table 1). Note that this is a compelling placebo test showing that training assignment is unrelated to pre-treatment secondary completion. Panel A in Table 3 shows that only about 5 and 7 percent of male and female training lottery losers, respectively, complete secondary school after the lottery. Male and female lottery winners are as likely as lottery losers to complete secondary school after the lottery. Note that the average age at which individuals take the SABER 11 exam is 20.2 years of age, and 38 percent of test-takers are 18 years or older. Participants in YIA, who are 18–25 years old, are thus well within the age range of those who take the SABER 11 exam, so older ages among participants are unlikely to be the reason why training has no effect on secondary education.

By contrast, the results in Panel B of Table 3 show that training participation increases long-term enrollment in tertiary education for training participants. About 15 percent of male lottery losers enroll in tertiary education after the training lottery. Male lottery winners are 3.7 percentage points (25 percent) more likely than losers to enroll in tertiary education up to 11 years after the lottery. Female lottery winners are 3.2 percentage points (28 percent) more likely to enroll in tertiary education than lottery losers 11 years after the lottery. The results are robust to the exclusion of baseline characteristics.¹⁹ On the other hand, a placebo test cannot reject the null of no pre-treatment differences between lottery winners and losers in tertiary enrollment (second-to-last row in Panel A of Table 1). This reinforces the causal interpretation of tertiary enrollment impact estimates.²⁰ These results contrast evidence from a randomized summer employment program for teenagers in New York City, in which participants were as likely as nonparticipants to enroll in college after the program (Gelber, Isen, and Kessler 2016).

19. We carried out robustness checks, leaving out all baseline characteristics and leaving out, one at a time, the characteristics that are significantly different between lottery winners and losers for either men or women at baseline (age, marital status, job tenure, and hours worked). The results from these checks, shown in [Online Appendix Table A1](#), are of similar magnitude and significance for both males and females, suggesting that differences in baseline characteristics do not drive the results.

20. In [Online Appendix Table A2](#), we adjust for multiple hypotheses testing by using the Anderson-adjusted false discovery rate (FDR) q -values. These q -values allow adjusting for the possibility of false discoveries (that is, failing to reject null hypotheses when they are false) when hypotheses are tested for various subgroups at once. In our context, we have six subgroups: male participants, female participants, male relatives of male participants, female relatives of male participants, male relatives of female participants, and female relatives of female participants. The results using the FDR q -values are less precise for tertiary schooling after YIA, but show similar general results.

Table 3
Training Impacts on Applicants' Formal Education Outcomes

	All Applicants		Males		Females		Male-Female Difference <i>p</i> -Value (7)
	Control Mean (1)	Participant Effect (2)	Control Mean (3)	Participant Effect (4)	Control Mean (5)	Participant Effect (6)	
Panel A: Secondary School Completion							
Completed secondary school after training	0.061	0.009 (0.008)	0.049	0.01 (0.011)	0.071	0.007 (0.012)	0.583
Panel B: Initial and Continued Tertiary Enrollment							
Tertiary enrollment after training	0.129	0.031*** (0.011)	0.146	0.037** (0.018)	0.115	0.032*** (0.015)	0.875
Observations	3,953		1,822		2,131		

Notes: The table reports control group means and differences between participants and controls among all applicants and separately by gender for educational outcomes that include secondary school completion (as measured by whether applicants take the secondary school SABER 11 test), and tertiary education initial and continued enrollment. The data for the secondary school outcomes come from the SABER 11 database. The data for the tertiary education outcomes from the SPADES database. Participant effect estimates are obtained from OLS regressions that include as additional control variables (not shown): gender, age, married, secondary completed before training, enrolled in tertiary before training, working according with the SISBEN census of the poor, salary, profit, formal employment, contract, tenure, days worked, hours worked, and training institution fixed effects. The last column reports the *p*-value of the test of difference in treatment effects between females and males. Robust standard errors are in parentheses. **p* < 0.1, ***p* < 0.05, ****p* < 0.01.

B. Impacts on Relatives' Secondary School Completion and Tertiary Enrollment

In this section, we examine whether training participation indirectly affected the educational outcomes of participants' relatives. If families share educational resources, training may ease household educational credit constraints due to increased participant incomes in the short term (Attanasio, Kugler, and Meghir 2011), and training participation may also benefit other family members' schooling. There may also be informational externalities, as other family members learn about new skills or about returns to schooling, both general and field-specific. We return to these potential mechanisms in Section V.

We find that relatives of lottery winners are more likely than relatives of lottery losers to complete secondary school in the long term. Relatives of female lottery winners are 1.7 percentage points (14 percent) more likely to complete secondary school after the training lottery, while relatives of male lottery winners as a whole do not experience any increase in secondary schooling (Table 4, Columns 2 and 4, Panel A, Row 1).²¹

Indirect effects on secondary school completion are larger for female relatives than male relatives of YIA participants (Table 4, Columns 2 and 4, Panel A, Rows 2 and 3). Male relatives of male lottery winners are not more likely to complete secondary schooling after training (Table 4, Column 2, Panel A, Row 2). By contrast, Row 3 of Panel A in Table 4 shows that secondary school completion rates among female relatives of male lottery winners are 2.4 percentage points (19 percent) more likely to complete secondary school. Female relatives of female lottery winners are 2.4 percentage points (21 percent) more likely to complete secondary school than female relatives of losers. The difference with female relatives of male lottery winners is significant at the 10 percent level. Importantly, the placebo tests in Panel B of Table 2 show that secondary school completion of neither female nor male relatives of lottery winners was higher before participation in the program, giving credibility to a causal interpretation of these results.²² By contrast, male relatives of either female or male lottery winners are not more likely to complete secondary school after YIA.²³

Training participation also appears to increase tertiary school enrollment of relatives of male participants, but not of relatives of female participants. Tertiary enrollment increases by two percentage points (22 percent) for relatives of male lottery winners, as compared to relatives of lottery losers (Table 4, Column 4, Panel B, Row 1), an effect driven by female relatives (Table 4, Column 6, Panel B, Row 2). Unfortunately, the placebo test for relatives of male lottery winners shows a preexisting difference in tertiary school enrollments, which means that we cannot interpret this post-training result causally (Row 4, Panel B, Table 2). On the other hand, when we separate female and male relatives, we find significant effects for female relatives post-training (Row 3, Panel B, Table 4), but insignificant results pre-training (Row 6, Panel B, Table 2). The

21. [Online Appendix Table A3](#) reports the results, excluding all baseline controls, and the results are unchanged.

22. We cluster standard errors by household for all specifications of household relatives' outcomes. In [Online Appendix Table A4](#), we also show significance adjusted for false discovery rate accounting for six different subgroups. The results for female relatives of both male and female participants are significant at conventional levels with either the naïve p -value or the FDR q -value.

23. We also tried estimating differential effects for relatives who were older or younger than the training participants, but found no differences, so we do not report them here.

Table 4
Impacts on Household Relatives' Formal Education Outcomes

	All Applicants		Males		Females		Male–Female Difference <i>p</i> -Value (7)
	Control Mean (1)	Participant Effect (2)	Control Mean (3)	Participant Effect (4)	Control Mean (5)	Participant Effect (6)	
Panel A: Secondary School Completion							
Completed secondary school after training	0.127	0.013** (0.006)	0.131	0.013 (0.009)	0.124	0.017** (0.008)	0.507
Completed secondary school after training (male relative)	0.134	0.003 (0.008)	0.133	0.001 (0.012)	0.136	0.008 (0.012)	0.643
Completed secondary school after training (female relative)	0.12	0.023*** (0.008)	0.128	0.024* (0.013)	0.115	0.024** (0.01)	0.769
Panel B: Initial and Continued Tertiary Enrollment							
Tertiary enrollment after training	0.086	0.01** (0.005)	0.089	0.02** (0.008)	0.084	0 (0.006)	0.035
Tertiary enrollment after training (male relative)	0.086	0.005 (0.007)	0.085	0.014 (0.01)	0.086	−0.006 (0.009)	0.099
Tertiary enrollment after training (female relative)	0.087	0.013* (0.007)	0.094	0.025** (0.012)	0.083	0.002 (0.009)	0.116
Observations	15,733		7,014		8,719		

Notes: The table reports control group means for formal education outcomes of applicants' relatives in the same household at baseline and differences in educational outcomes between relatives of participants and of controls among all applicants and separately by applicants' gender. Educational outcomes are secondary school completion (as measured by whether applicants take the secondary school test, SABER 11 Test), and tertiary education initial and continued enrollment. The data for the secondary school outcomes come from the SABER 11 database. The data for the tertiary education outcomes data come from the SPADIES database. Participant effect estimates are obtained from OLS regressions that include as additional control variables (not shown): applicants' gender, age, married dummy, secondary completed before training, enrolled in tertiary before training, working according with the SISBEN census of the poor, salary, profit, formal employment, contract, tenure, days worked, hours worked, and training institution fixed effects. The last column reports the *p*-value of the test of difference in treatment effects between females and males. Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

results show an overall effect for female relatives of 1.3 percentage points (15 percent) driven by the effect of 2.5 percentage points (27 percent) for female relatives of male participants. Contrary to the finding for male relatives, the placebo for female relatives shows no preexisting difference in Table 4.

C. Impacts on Long-Term Labor Market Outcomes of Participants and Relatives

We examine whether the short-term effects found in Attanasio, Kugler, and Meghir (2011) for labor market outcomes and in Attanasio et al. (2017) for the medium term are also evident in our data up to eight years after random assignment.

In Table 5, we replicate short-term training impact estimates on labor market outcomes using the Attanasio, Kugler, and Meghir (2011) one-year survey follow-up and report them alongside the long-term effects estimated using administrative data.²⁴ Over this period, the training effect on formal employment for males halves. One year after the lottery, male lottery winners are 7.9 percentage points (32 percent) more likely than losers to be in formal employment. Three to eight years after the lottery, they are four percentage points (5.1 percent) more likely than losers to be formally employed (Table 5, Columns 2 and 4, Panel B, Row 1). These short- and long-term formal employment impact estimates for males are significantly different from each other at the 5 percent level. The training effect on formal employment for females is higher in magnitude for women in the long term. One year after the lottery, female lottery winners are 6.2 percentage points (43 percent) more likely than losers to be formally employed. Three to eight years after the lottery, female lottery winners are 6.5 percentage points (11.3 percent) more likely than losers to be formally employed (Table 5, Columns 2 and 4, Panel C, Row 1).

Earnings impact estimates among males become insignificant three to eight years out. The earnings gain estimate for male lottery winners goes from significant gains of \$1.07/day one year after the lottery to a statistically insignificant \$0.10 three to eight years out (Table 5, Columns 2 and 4, Panel B, Row 2). However, the short- and long-term daily earnings impact difference among males is not statistically significant (Column 6). The earnings impact of female trainees is considerably more persistent than that of males. The earnings gain estimate for female lottery winners goes from \$0.66/day one year after the lottery to \$0.26/day (Table 5, Columns 2 and 4, Panel C, Row 2). The difference in short- and long-term daily earnings impacts among females is not statistically significant (Column 5).²⁵ However, even the lower earnings in the long term represent one-third of the female earnings effect in the short run, thus showing much more persistence than for men, for whom the long-term effect is only one-tenth of the short-term effect.²⁶ The differences between men and women in the long term could be due to the fact that

24. We cluster standard errors by household for all specifications of household relatives' outcomes. In [Online Appendix Tables A2 and A4](#), we also show significance adjusted for false discovery rate accounting for six different subgroups. The results for female relatives of both male and female participants are significant at conventional levels with either the naïve p -value or the FDR q -value.

25. Note that our imputation approach implies that impact estimates likely represent a lower bound on daily total earnings (see Section III for discussion). However, even with these lower bound estimates, 63 percent of the female earnings effect persists in the long term.

26. [Online Appendix Table A5](#) includes the FDR q -values and shows that the results are similar when we allow for false discoveries due to multiple hypothesis testing.

Table 5*Training Impacts on Applicants' Short- and Long-Term Labor Market Outcomes*

	Short Run		Long Run		
	Control Mean (1)	Participant Effect (2)	Control Mean (3)	Participant Effect (4)	Difference (<i>p</i> -Value χ^2) (5)
Panel A: Full Sample					
Formally employed	0.188	0.071*** (0.013)	0.664	0.057*** (0.014)	−0.014 (0.450)
Daily earnings	1.8	0.852*** (0.142)	8.97	0.218** (0.092)	−0.634*** (0.000)
Observations	3,799		3,799		
Panel B: Male					
Formally employed	0.245	0.078*** (0.023)	0.781	0.045** (0.02)	−0.033 (0.211)
Daily earnings	2.35	1.13*** (0.252)	10.6	0.195 (0.143)	−0.935*** (0.000)
Observations	1,682		1,682		
Panel C: Female					
Formally employed	0.145	0.064*** (0.017)	0.575	0.07*** (0.021)	0.006 (0.797)
Daily earnings	1.37	0.658*** (0.168)	7.71	0.264** (0.123)	−0.394** (0.026)
Observations	2,117		2,117		

Notes: The table reports control group means and differences between participants and controls among all applicants and separately by gender in the probability of working in the formal sector, and daily earnings in the short and long run. Short-run estimates are obtained from the one-year follow-up survey (2006). Formal employment in the short run is defined as whether applicants are covered by health insurance, injury compensation, pensions, or family subsidies. Earnings in the short run are salaries and wages earned in the main job held during the year after having finished training for salaried workers and include both formal and informal earnings. For long-run outcomes, formal employment is whether the applicant appears in the SISPRO Social Security database in 2013 as contributing to the pension fund. Long-run earnings are formal earnings for those in formal employment. For those not in a formal employment, we imputed informal earnings from the 2008 and 2010 Colombian household surveys for nationally representative household surveys, using the information from those who would have been between 18 and 25 years of age in 2005, who are in the lowest two deciles of the income distribution, and who live in the eight cities in which training was provided. We then imputed the average expected earnings of the nonformally employed by multiplying the likelihood of unemployment times the likelihood of informal employment times average informal-sector earnings for the group of workers with the same characteristics as YIA applicants workers from the household surveys. Earnings expressed in 2013 USD (1 USD=1,869.1 Colombian pesos). Column 5 reports the *p*-value of the differences in treatment effects between short-run and long-run effects (Columns 2 and 4). Robust standard errors are in parentheses. **p*<0.1, ***p*<0.05, ****p*<0.01.

Table 6
Impacts on Household Relatives' Short- and Long-Term Labor Market Outcomes

	Short Run		Long Run		Difference (<i>p</i> -Value χ^2) (5)
	Control Mean (1)	Participant Effect (2)	Control Mean (3)	Participant Effect (4)	
Panel A: Full Sample					
Formally employed	0.218	0.004* (0.002)	0.252	0.014 (0.009)	0.01 (0.259)
Daily earnings	1.02	0.035 (0.031)	8.87	0.034 (0.094)	−0.001 (0.991)
Observations	10,184		10,184		
Panel B: Male					
Formally employed	0.265	−0.001 (0.003)	0.236	−0.001 (0.013)	0 (0.960)
Daily earnings	1.29	0.014 (0.05)	8.81	0.103 (0.144)	0.089 (0.560)
Observations	4,849		4,849		
Panel C: Female					
Formally employed	0.178	0.008** (0.004)	0.266	0.026** (0.013)	0.018 (0.148)
Daily earnings	0.779	0.069* (0.037)	8.92	−0.072 (0.132)	−0.141 (0.313)
Observations	5,335		5,335		

Notes: The table reports control group means and differences between household relatives of participants and those of controls among all applicants, and separately by applicants' gender for relatives labor market outcomes in the short and long run. Short-run estimates are obtained from the one-year follow-up survey (2006). Formal employment in the short run is defined as whether applicants are covered by health insurance, injury compensation, pensions, or family subsidies. Earnings in the short run are salaries and wages earned in the main job held during the year after having finished training for salaried workers and include both formal and informal earnings. For long-run outcomes, formal employment is whether the applicant appears in the SISPRO Social Security database in 2013 as contributing to the pension fund. Long-run earnings are formal earnings for those in formal employment. For those not in a formal employment, we imputed informal earnings from the 2008 and 2010 Colombian household surveys for nationally representative household surveys, using the information from those who would have been between 18 and 25 years of age in 2005, who are in the lowest two deciles of the income distribution, and who live in the eight cities in which training was provided. We then imputed the average expected earnings of the nonformally employed by multiplying the likelihood of unemployment times the likelihood of informal employment times average informal-sector earnings for the group of workers with the same characteristics as YIA applicants workers from the household surveys. Earnings expressed in 2013 USD (1 USD = 1,869.1 Colombian pesos). Column 5 reports the *p*-value of the differences in treatment effects between short-run and long-run effects (Columns 2 and 4). Robust standard errors are in parentheses. **p* < 0.1, ***p* < 0.05, ****p* < 0.01.

men are still enrolled in the formal education system, most likely at the tertiary level, while women are closer to fully realizing their educational gains in the labor market.

Overall, we interpret the smaller long-term effects on employment and earnings, as compared to the short-run effects, as being due to the control group catching up. Comparing Columns 1 and 3 in Table 5 shows that the control group means increase over time. This catch-up of untreated individuals over time is likely due to control group individuals being less likely to be in school during that same time and accumulating experience. Thus, in the short term, the program helps treated individuals leapfrog with respect to the control group. Then the control group improves and closes the gap with the treated individuals.

We find no evidence of short- or long-term formal employment or earnings effects on relatives of male participants (Table 6, Columns 2 and 4, Panel B, Rows 1 and 2). We do, however, find some evidence of positive spillovers on short-term formal employment and earnings and on long-term formal employment among relatives of female participants (Table 6, Columns 2 and 4, Panel C, Rows 1 and 2). A possibility is that relatives of male participants, who appeared to get more secondary education, are still going through the formal education system, and the effects have not yet materialized in the form of better labor market outcomes. Aside from the long-term employment effect for relatives of female trainees, none of the other coefficients for the long-term effects in Table 6 are significant.

V. Potential Channels through Which Training Increases Formal Schooling

In this section, we examine potential channels through which vocational training increases subsequent formal schooling of beneficiaries and other household members. We focus on three potential channels: the relaxation of household credit constraints for education investments; information about general returns to education investments or attainment of general education; and acquisition of actual occupation-specific skills or information about occupation-specific returns.

A. Credit Constraints

Results in Attanasio, Kugler, and Meghir (2011) indicate that one year after the lottery, training increased the probability of formal paid employment by seven percentage points and earnings by 20 percent among female trainees. Among men, training had no short-term impact on earnings but had positive impacts on formal employment. Our results in the previous section show similar positive short- and long-term employment impacts of YIA for both men and women using administrative data.

Participants and their relatives thus may have increased educational attainment after winning the YIA lottery, to the extent that stipends during training and higher earnings following training mitigated educational credit constraints. For example, some educational outlays, such as tertiary school tuition, are large, and households may not have enough savings at a point in time to pay for all of them. Additional earnings resulting from training may help participants and relatives cover these fixed and sunk costs of further education acquisition. We first test for the presence of credit constraints. Then we

test whether those with higher ability (as measured by the secondary school graduation test scores) as well as those with higher earnings post-training had a higher likelihood of pursuing more formal education subsequent to vocational training.

We test for the possible presence of borrowing constraints using the strategy of Cameron and Taber (2004). We specify a Mincer equation to estimate the returns to education:

$$Earnings_{ickt} = \phi + \rho Education_{it} + \beta X_{ickt} + \kappa_k + \tau_t + \omega_c + \varepsilon_{ickt},$$

where the dependent variable is the earnings of individual i , living in city c working in sector k in year t . On the right-hand side, the variable $Education_{it}$ captures the years of schooling, X_{ickt} is a vector that includes other standard variables present in a Mincerian equation (namely, potential experience, marital status, and gender), and κ_k , τ_t , and ω_c are fixed effects for sector, year, and city.

Cameron and Taber (2004) propose to test for credit constraints by estimating the above regression using an instrumental variables (IV) approach relying on two separate variables, direct costs and indirect costs, to instrument for educational attainment. This approach then tests whether the returns are higher when using direct costs as the IV than when using indirect costs as the IV. The intuition behind the Cameron and Taber (2004) approach is that in the presence of credit constraints, schooling decisions will be more sensitive to changes in the direct costs of schooling than the indirect costs for those who are credit constrained. Credit-constrained individuals will be less responsive to a higher monetary return to schooling, as they lack the liquidity to make the investment. To the extent that IV estimates using direct costs as a source of exogenous variation are local to credit-constrained individuals, this IV approach would produce high estimated returns to schooling. By contrast, instrumenting schooling with opportunity costs, mainly foregone income from becoming a student, should lead to smaller IV estimates of the return to schooling, since indirect costs affect credit-constrained and credit-unconstrained individuals equally. As in previous studies, this approach relies on stringent exclusion restrictions that require both direct and indirect costs to not affect earnings through any channel other than through years of education.

To implement the Cameron and Taber (2004) approach, we use the following as sources of variation in direct costs: commuting distance to the nearest college, whether the nearest college is private, and the interaction of commuting distance and private status. We estimate the commuting distance as the shortest travel distance from the person's household address at baseline to the nearest college. We do find that Colombian students in this socioeconomic group, like students in the United States, tend to go to the nearest college, so that distance matters and imposes costs. Table A6 in the [Online Appendix](#) shows that among applicants who attend tertiary education, the average distance to the college in which they enroll is less than the median distance to all available colleges. We measure these distances for each individual using georeferencing and relying on Google maps. Distance to college and whether the nearest college is private likely satisfy the exclusion restrictions for various reasons. First, we are determining distance to college based on applicants' residential location at baseline, which is exogenous to treatment assignment. Thus, even if applicants moved to attend college as a result of treatment, distance from residential location at baseline would satisfy the exclusion restriction. Second, most students attend college in the city in which they reside, and all colleges in Colombia are geographically clustered in the main

metropolitan areas. Since these metropolitan areas represent well-defined local labor markets, conditional on ability, distance is not correlated with labor market opportunities within the metropolitan area. Third, all of the colleges that applicants in the sample attended were also established before program participation, such that the distances and whether the nearest college was private were determined before program participation.

As a source of indirect costs, we use the opportunity cost of college attendance, measured by gender-specific average wages of high school graduates in each metropolitan area, which corresponds to a local labor market, estimated at baseline using the 2005 SISBEN census of the poor. Since the SISBEN census was conducted prior to program participation, these are *ex ante* returns. High prevailing earnings in the local labor market imply high foregone earnings of college attendance. The major concern in using this variable as an instrument, as Cameron and Taber (2004) note, is that local labor market conditions at baseline are very likely correlated with local labor market conditions later, when students make college attendance decisions.²⁷ After estimating returns-to-schooling coefficients using both IV specifications, we test if the estimate of ρ is greater when the instrument is the direct cost rather than the indirect cost.

Table 7 presents results from this test of the existence of credit constraints. Columns 1–4 in Table 7 show the results for the full sample. Both instruments are relevant, as measured by the corresponding first-stage *F*-statistic (Table 6, Columns 2 and 3). For the full sample, the results indicate that there is no significant difference between the return-to-schooling coefficients estimated with the direct and indirect cost instruments (Table 6, Column 4), showing no indication of credit constraints. However, when splitting the sample between men and women, the results show evidence consistent with women being credit constrained. Columns 6 and 7 show that while the opportunity cost instrument is relevant for men, the first stage for the direct cost instrument is not relevant. By contrast, Columns 10 and 11 show that both direct and indirect costs are relevant instruments for women. Moreover, the results show that the returns estimated using direct commuting costs are higher than the returns estimated using indirect costs and that this difference is significant (Table 6, Column 12). Thus, we find evidence consistent with the presence of credit constraints for college attendance for women, implying that the relaxation of credit constraints could be driving the formal education effects documented in the previous section for women participants, as well as the spillovers on relatives of women participants. Women, indeed, appear to be more credit constrained before the program, as the baseline characteristics in Table 1 show women earning lower wages than men and having a lower likelihood of having formal employment.

We then test if those more likely to respond to credit constraints are also more likely to increase formal education subsequent to training. [Online Appendix Table A7](#) shows the results of a model that includes the test score for secondary school graduation (as a proxy of ability) and an interaction of the YIA lottery winner dummy with the test score variable.²⁸ If credit constraints are relaxed, those who have greater academic ability will

27. One way to circumvent this issue would be to include as an additional control current local labor market conditions in the earnings equation. However, we do not have a good measure of local labor market conditions that can be used.

28. For those who did not take the exam and for whom we have no score, we impute the scores of the 25th percentile by gender. The assumption is that those who did not take the exam would have been among the lower-performing students.

Table 7
Test of Liquidity Constraints: Returns to Education Using IVs for Direct + Indirect Costs

	Full Sample				Males				Females			
	OLS (1)	IV (Direct Cost) (2)	IV (Indirect Cost) (3)	Direct – Indirect (χ^2 -Test) (4)	OLS (5)	IV (Direct Cost) (6)	IV (Indirect Cost) (7)	Direct – Indirect (χ^2 test) (8)	OLS (9)	IV (Direct Cost) (10)	IV (Indirect Cost) (11)	Direct – Indirect (χ^2 -Test) (12)
Panel A: Outcome Is $\sin h - 1$ (Baseline Earnings)												
Years of Schooling	-0.693 (0.462)	-14.7 (21.8)	-9.22 (6.44)	0.088	-1.17* (0.645)	-1.85 (9.13)	-10.2 (8.57)	4.39	-0.091 (0.662)	47* (27.5)	15.2* (8.96)	1.74
Obs.	3,473	3,473	3,473	3,473	1,581	1,581	1,581		1,892	1,892	1,892	
R^2	0.06				0.068		0.067		0.058			
Panel B: First Stage: Outcome is Years of Schooling												
Distance		-0.0005* (0)				-0.0003 (0)				-0.0009* (0.001)		
Private		-0.0009 (0.009)				-0.017 (0.015)				0.009 (0.011)		
Distance $\times$ Private		0.002 (0.001)				0.004* (0.003)*				0 (0.001)		
Foregone earnings in reference group			-0.002*** (0)				-0.001*** (0)				-0.006*** (0.002)	
F-test		2.09	3.48			1.81	2.56			1.99	2.8	

Notes: The table reports results from Mincerian equations for applicants' years of schooling at baseline. OLS regressions use as additional controls (not shown): baseline age, age squared, gender, and metropolitan area. In IV specifications for direct college costs, years of schooling is instrumented by (Euclidean) distance to the nearest college, whether the nearest college is private, and an interaction of the two, with the same controls as OLS. In IV specifications for indirect college costs, years of schooling is instrumented by foregone earnings, which for applicant i are average earnings at baseline for applicants of the same gender and years of education as i , excluding earnings from applicant i . Baseline earnings includes zeros, and are transformed using the inverse hyperbolic sine. IV regressions are used for the direct and indirect costs; both use the same controls as the OLS regression. The last column reports the Chi test of difference in treatment effects between direct and indirect cost. Robust standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

be more likely to enroll in tertiary education.²⁹ The results show little evidence that men are more likely to enroll in tertiary education after training if they have higher scores. On the other hand, there is evidence that women with higher scores do enroll more in tertiary schooling after participating in YIA. [Online Appendix Table A8](#) shows results of models with interactions of the YIA indicator with an indicator of whether individuals earned above the mean two years after YIA. The results do not show much evidence that those who earned more after YIA went on to pursue tertiary education.

Credit constraints would also predict that if individuals are indeed credit constrained, any tertiary enrollment effects should concentrate on enrollment in low-cost institutions. [Online Appendix Table A9](#) shows evidence that men are more likely to enroll in high-cost private universities, while women are not, and the differences between men and women are marginally statistically significant. These various pieces of evidence, together with the Cameron–Taber test, point to the relevance of credit constraints to explain why training increased subsequent tertiary education for women.

B. Learning about Returns to General Skills or Acquiring General Skills

Another reason why vocational training may increase subsequent formal education is through the acquisition of general information during training about the rate of return to formal education. There is evidence in other contexts (for example, Jensen 2010) that when students in secondary education have better information about the benefits of tertiary education, they respond by staying in school longer. Also, formal education effects of training may arise because trainees acquire general skills, which make it easier to continue studying.

In Table 8, we test directly whether the tertiary enrollment effects of participation in YIA are greater when ex ante returns to tertiary education are higher. The variation in returns to schooling comes from estimating Mincerian wage equations of the return to a university education separately by gender and city, controlling for education, experience, and a quadratic in experience. For this approach, we use data from the SISBEN 2005 data so that we can get ex ante returns to a university education. Then we reestimate training impacts on tertiary education enrollment in a regression model as before, but also include ex ante returns to a university education and the YIA indicator, interacted with the returns to a university education. We do not find evidence that YIA drives either men or women to acquire more formal education when they live in a city with higher general returns to schooling.³⁰ Note, however, that these results are suggestive,

29. A caveat to this test is that, as long as there is rationing of slots at institutions of tertiary education by academic ability, we would expect that students with higher ability would be able to enroll in tertiary education more easily, even in the absence of credit constraints. However, in Colombia, most rationing of tertiary education slots by academic ability takes place in public flagship universities for which there are no financial constraints to attendance, since low-income students receive full subsidies conditional on being admitted. There is limited rationing by academic ability among a handful of very selective and expensive private institutions, which none of the applicants in our sample attend. The universities that applicants in the sample attend are largely private or public open-enrollment institutions, suggesting that this test for credit constraints is indeed helpful in the Colombian context.

30. There is only a marginally significantly negative coefficient on the interaction term for men in terms of their tertiary enrollment and a negative impact on completed secondary schooling significant at the 5 percent level.

Table 8
Training Impacts on Applicants' Formal Education Outcomes, by Ex Ante Returns to Education

	All			Males			Females			Male-Female Difference <i>p</i> -Value (10)
	Control Mean (1)	Participant Effect (2)	Interaction (3)	Control Mean (4)	Participant Effect (5)	Interaction (6)	Control Mean (7)	Participant Effect (8)	Interaction (9)	
Panel A: All Applicants										
Completed secondary school after training	0.061	0.098** (0.042)	-0.093** (0.044)	0.049	0.092 (0.057)	-0.091 (0.065)	0.071	0.076 (0.076)	-0.069 (0.074)	0.958
Tertiary enrollment after YIA	0.129	0.1 (0.062)	-0.073 (0.063)	0.146	0.195** (0.087)	-0.178* (0.094)	0.115	0.008 (0.1)	0.024 (0.098)	0.066
Observations	3,953			1,822			2,131			
Panel B: All Relatives										
Completed secondary school after training	0.125	0.009 (0.031)	0.004 (0.032)	0.128	0.009 (0.044)	0.007 (0.048)	0.123	0.026 (0.051)	-0.01 (0.049)	0.369
Tertiary enrollment after YIA	0.086	0.035 (0.029)	-0.024 (0.029)	0.088	0.035 (0.043)	-0.011 (0.047)	0.083	-0.031 (0.043)	0.03 (0.041)	0.988
Observations	15,247			6,828			8,419			

Notes: The table reports training impacts on formal education outcomes of participants and relatives by ex ante returns to education. These returns are obtained from the SISBEN census of the poor from separate Mincerian equations of the inverse hyperbolic sine of earnings on years of schooling, age, and age squared, estimated separately by gender and metropolitan area. The last column reports the *p*-value of the test of difference in treatment effects between females and males. Robust standard errors are in parentheses for applicants, and household clustered standard errors are in parentheses for relatives. **p* < 0.1, ***p* < 0.05, ****p* < 0.01.

since in an ideal test we would be able to discern if the individual knows or obtains information on these *ex ante* returns.

Alternatively, it could be that YIA boosts general skills for individuals and that this allows them to pursue additional formal education more easily. In particular, we explore whether YIA led individuals to obtain higher test scores in the secondary school exam, which would also allow them easier access to tertiary education. Table 9 shows estimates of an ordinary least squares (OLS) regression of test scores on a YIA indicator. The results in Panel A show no effect of YIA on test scores for either male or female participants. Thus, these results indicate that YIA is not improving general skills for either male or female participants. Panel B shows no effects on test scores of relatives. Thus, there is also no evidence that participants or relatives are improving their general skills, as measured by test scores, as a result of YIA.

C. Learning about Field-Specific Returns or Acquiring Field-Specific Skills

An alternative informational externality may arise if participants learn about requirements and field-specific returns as they go through their training coursework and on-the-job apprenticeships.³¹ The classroom training and apprenticeships were fairly narrowly defined, potentially enabling individuals to acquire field-specific knowledge. Alternatively, individuals may be learning skills that will be useful for continued study in a specific field. We find evidence consistent with this informational channel for men but not for women.

Table 10 reports transition probabilities for the full sample for men and for women. The transition probabilities show the share of individuals who studied training courses in particular fields and went on to pursue a college degree in different fields. These transition probabilities suggest stickiness in fields of study. For example, the one in the natural science diagonal indicates that 100 percent of those who pursued natural science training courses (for example, clinical lab assistant or environmental assistant) and subsequently enrolled in tertiary education did so in the fields of natural sciences and math. Thirty percent of those who took economics and business training courses and subsequently enrolled in tertiary education did so in economics or business-related majors. Forty percent of those who took a training course in construction (for example, construction operator or molding and foundry worker) and subsequently enrolled in tertiary education did so in engineering majors. Forty percent of those who trained in health and education and subsequently enrolled in tertiary education did so in health or education majors. When transition probabilities are estimated for men and women separately, the probabilities along the diagonal are greater for men than for women, with the exception of economics and business. We find that 64 percent of men who pursued training in construction and subsequently enrolled in tertiary education do so in engineering majors. Also, 50 percent of men who take a course in health fields and subsequently enroll in tertiary education do so in health and education majors. Finally, 100 percent of the students who undertake YIA courses in natural sciences and subsequently enroll in natural science majors in college are men. Thus, the evidence in Table 9 shows

31. For example, Hastings, Neilson, and Zimmerman (2013) and Kirkeboen, Leuven, and Mogstad (2016) show that returns to field of study vary significantly more than returns to college quality.

Table 9
Impacts on Secondary School Exit Exam Scores of Applicants and Relatives

	All Applicants		Males		Females		Male-Female Difference <i>p</i> -Value (7)
	Control Mean (SD) (1)	Participant Effect (SD) (2)	Control Mean (SD) (3)	Participant Effect (SD) (4)	Control Mean (SD) (5)	Participant Effect (SD) (6)	
Panel A: All Applicants							
Secondary exit score after training	42.7	-7.55 (6.9)	45.7	-8.37 (10.7)	40.6	-10.7 (12)	0.923
Observations	167		72		95		
Panel B: All Relatives							
Secondary exit score after training	43.7	-1.86 (3.47)	43.2	-0.341 (5.83)	44.1	1.81 (5.32)	0.760
Secondary exit score after training (male relative)	37	1.26 (6.02)	35.8	-2.19 (13.9)	38.1	-9.6 (12.1)	0.193
Secondary exit score after training (female relative)	51	-5.38 (5.83)	53.1	20.9 (31.1)	49.7	6.65 (8.42)	0.188
Observations	201		84		117		

Notes: The table reports training impacts on secondary school exit exam scores, measured in percentiles. Baseline controls (not shown) include gender, age, married, secondary completed before training, enrolled in tertiary before training, employment status from the SISBEN census of the poor, salary, profit, formal employment, contract, tenure, days worked, hours worked, and training institution fixed effects. The last column reports the *p*-value of the test of difference in treatment effects between females and males. Columns 2, 5, and 8 report estimates from OLS regressions in the sample of applicants and relatives who took the SABER 11 test. Columns 3, 6, and 9 report estimates from tobit models with a lower-level censoring point at the 25th percentile of the corresponding distribution. Missing test scores or with scores below the 25th percentile are imputed at the 25th percentile score, which is 21 points in the applicant sample and 18 points in the household relatives sample. Robust standard errors are in parentheses. **p*<0.1, ***p*<0.05, ****p*<0.01.

Table 10
Enrollment in Tertiary Education's Knowledge Domains by Vocational Field of Training (Percent)

Training Field	Agricultural and Mining	Arts	Economics and Business	Engineering	Health and Education	Natural Sciences and Math	Social Sciences	Total General
Full sample:	10	4	30	40	3	4	16	100
Agricultural and mining	0	0	0	100	0	0	0	100
Arts	0	0	33	33	33	0	0	100
Economics and business	2	5	3	43	8	2	10	100
Construction and manufacturing	0	5	35	40	5	5	10	100
Health and education	0	0	20	20	40	0	20	100
Natural sciences	0	0	0	0	0	100	0	100
Social sciences	0	0	100	0	0	0	0	100
Males:	2	6	22	51	8	2	10	100
Art	0	0	0	100	0	0	0	100
Business, finance, and sales	3	6	21	52	6	0	12	100
Construction and manufacturing	0	9	27	64	0	0	0	100
Health	0	0	0	25	50	0	25	100
Natural sciences	0	0	0	0	0	100	0	100
Social sciences	0	0	100	0	0	0	0	100
Females:	0	2	41	27	12	7	10	100
Agricultural and mining	0	0	0	100	0	0	0	100
Art	0	0	50	0	50	0	0	100
Business, finance, and sales	0	4	41	33	11	4	7	100
Construction and manufacturing	0	0	44	11	11	11	22	100
Health	0	0	100	0	0	0	0	100
Natural sciences	0	0	0	0	0	100	0	100

Notes: For the 780 applicants who enrolled in tertiary education after training, the table reports the percent that chose a particular field of study given their training field of study during the YIA program.

that a high proportion of male trainees who pursue tertiary education do so in a field related to the field in which they trained. This is consistent with either men learning about returns to field-specific skills or acquiring the field-specific skills that allow them to continue studying in that field.

Overall, we find evidence that training increases subsequent tertiary education for women, probably by the relaxation of credit constraints. In contrast, training increases tertiary education for men probably by allowing male participants to learn about specific field skills or specific returns to these skills while they train.

VI. Welfare Analysis

To illustrate the welfare implications of the impacts of vocational training on subsequent formal education, we conduct a cost–benefit analysis under three different scenarios: (i) only accounting for labor market impacts of participants accruing from vocational training participation, (ii) accounting also for increased future earnings of participants due to increased completed tertiary education, and (iii) also accounting for increased future earnings of relatives due to increased secondary education.

We estimate the costs and benefits of the program and calculate internal rates of return separately for women and men.³² There are three sources of costs. The first are direct program costs. The second is the marginal cost of public funds raised through taxation. The third is the loss of job tenure associated with program participation. Direct program costs are \$770 per participant. We impute the marginal cost of public funds to be 40 cents per dollar raised through taxation (Barrios, Pycroft, and Saveyn 2013). The loss of tenure is 1.43 months, which evaluated at baseline earnings is \$64 per participant (Attanasio, Kugler, and Meghir 2011).

There are three main sources of program benefits: direct benefits to participants, educational spillovers on participants, and additional educational spillovers on relatives of participants. Benefits to participants are only those from increased total earnings (formal plus informal, Table 11). We do not account for nonwage benefits of increased formal-sector employment or other potential benefits, such as delayed childbearing or reduced program participation in other government programs (for example, Bettinger et al. 2018).

To compute the net present value (NPV) of direct benefits on participants, we assume a 40-year work horizon. We project annual total earnings using the annual earnings estimates from Tables 4 and 5. Specifically, we run a linear regression of annual total earnings for 2006–2013 on a linear calendar-year term, separately by gender and lottery status.³³ Figure A1 in the [Online Appendix](#) shows the projected earnings profiles of lottery winners and losers, separately by gender. Thin red lines describe direct earnings

32. Throughout the analysis, for each source of cost and revenue, we compute the net present values (NPVs), converting into U.S. dollars using the year-specific exchange rate (annual average) between U.S. dollars and Colombian pesos from the Colombian Central Bank, deflating nominal costs back to real value in base year (1995) using the U.S. CPI change between the base year and accrual of costs (or revenue), and taking the present value of the cost and revenue stream. We express the NPV in U.S. dollars for the year of analysis (2014) using the U.S. CPI change between the year of analysis and the base year.

33. We do not observe earnings for 2007. We estimate 2007 earnings from an interpolation of 2006 and 2008 earnings, separately by gender and lottery status.

Table 11
Program Costs and Internal Rates of Return

	Female Applicants (1)	Male Applicants (2)
Panel A: Program Cost (Per participant)		
Direct program cost, USD		770
Marginal cost of public funds, USD (40 cents/dollar)		308
Lost of tenure cost		64
Total cost (per participant)		1,142
Panel B: Internal Rates of Return		
Direct earnings effect on participants	0.221	0.102
Direct effect + education spillover on participants	0.235	0.205
Direct effect + education spillover on participants and relatives	0.241	0.255

Notes: The table reports costs of the training program and internal rates of return under three different scenarios. Estimates that only account for direct earnings benefits of participants use total annual earnings in 2013 (formal and informal) for participants and controls, projected linearly after 2013, separately for each treatment group and gender, and then assume 10 percent of annual depreciation. See notes to Table 5 for additional details on total earnings estimation. These earnings projections are then deflated back to real values in base year (2005) using the U.S. consumer price index (CPI), which for years after 2014 is assumed to be that of 2014. The net present value is then inflated forward to year of analysis (2014) using the ratio of the U.S. CPI in the year of analysis to that in the base year. Estimates that account for additional formal education achievement among applicants use the program impacts on the three-year continued tertiary enrollment, full sample separately by gender ([Online Appendix Table A2](#)), and assume a 10 percent annual return to an additional year of education.

profiles of participants without educational spillovers, while dashed black lines show the earnings for nonlottery winners. The earnings of lottery winners are notably higher than those of lottery losers at every age for women, but not for men. The IRRs that account for only the benefits from direct participant earnings are 22.1 percent for females and 10.2 percent for males (Table 11, Panel B, Row 2). Since 54.8 percent of our sample is female, we estimate an IRR for the full sample of 16.75 percent, very similar to the 16 percent IRR reported by Attanasio et al. (2017).

To account for the additional earnings benefits of increased tertiary enrollment and retention among participants, we assume a 10 percent return to an additional year of tertiary education and multiply these gains by the gender-specific program impact on the three-year tertiary education retention rate. We assume that the educational spillover benefits begin to accrue eight years after vocational training completion.³⁴ Thick continuous black lines in [Online Appendix Figure A1](#) describe earnings profiles on

34. The ex ante Mincerian returns to schooling estimated from the 2005 SISBEN census of the poor are between 8 and 12 percent, depending on gender and metropolitan program area, so we took the median value for

participants including education spillovers. The earnings of lottery winners once accounting for education spillovers are notably higher than those of lottery losers at every age for both women and men. The IRRs in which benefits include direct earnings effects on participants plus additional earnings due to the impacts of vocational training on subsequent formal education are 23.5 percent for women and 20.5 percent for men (Table 11, Panel B, Row 2).

We also estimate the benefits to nonparticipants, as we found additional secondary school completion rates among relatives of participants. We value the educational benefits on relatives using a 10 percent rate of return to an additional year education from baseline earnings of relatives, which comes from the same Mincerian equation we estimated for Tables 7 and 8 using the SISBEN 2005 data. [Online Appendix Figure A2](#) shows the earnings for relatives of lottery winners (continuous lines) and losers (dashed lines) and shows that the earnings of relatives of lottery winners exceed those of lottery losers at every age. The IRRs when we account for educational spillovers on relatives in addition to direct benefits and subsequent formal education of participants are 24.1 percent for women and 25.5 percent for men.

Explicitly accounting for the impact of vocational training on formal education improves the assessment of the program's net welfare benefits, particularly for men.³⁵ For both women and men, IRRs exceed the return on Colombian sovereign bonds of 7–9 percent over the period. This analysis illustrates the welfare implications of previously undocumented impacts of vocational training on formal education, as well as possible welfare implications of household spillovers. Failing to account for these additional effects may substantially underestimate the social desirability of vocational programs like Colombia's YIA.

VII. Conclusion

We analyze the long-term direct and spillover effects of a randomized vocational training program for disadvantaged youth implemented at scale in Colombia. We provide the first evidence using an experimental research design that vocational training generates subsequent formal educational investments. We find evidence that vocational training increases tertiary education of participants and that training participation has spillover effects on the secondary education of relatives of participants.

We then attempt to disentangle the channels through which training increases subsequent formal education. We find evidence consistent with the relaxation of credit constraints being the primary driver of subsequent university investments for women.

these calculations. The specific formula to account for educational spillovers is a Mincerian parallel shift in which, beginning in 2014, total predicted earnings of lottery winners with the educational spillovers are equal to predicted earnings of winners without spillovers times $[1 + \text{Prob}(3\text{yr retention}) \times (1 + 0.1)^3]$. The assumption that educational spillover benefit begin to accrue only after eight years is conservative and, if anything, would lead to understating the IRR for the case of spillovers to the extent that we do not account for possible earnings benefits from these educational spillovers in prior years after vocational school completion.

35. This calculation assumes that there are no negative spillovers to nonparticipants, for instance, in the form of job displacement effects (for example, Crépon et al. 2013). Attanasio, Kugler, and Meghir (2011) present evidence that the program did not have differential effects in tight and weak labor markets and, thus, argue that this evidence does not show displacement for nonparticipants.

On the other hand, we find that the acquisition of specific skills or information about the returns to specific skills is likely driving the increased tertiary education of men after training.

We estimate costs and benefits of the program and calculate internal rates of return separately for women and men. These calculations highlight the importance of accounting for additional human capital investments downstream as part of the benefits when conducting welfare analysis. Program IRRs, in which benefits stem solely from direct earnings effects on participants, are 22.1 percent for women and 10.2 percent for men. Once we account for the impact of vocational training on the subsequent formal education of participants and their relatives, the IRRs are 24.1 percent for women and 25.5 percent for men.

Taken together, our results indicate that a vocational training program such as Colombia's YIA that was designed to combine classroom training, apprenticeships, and noncognitive skills training, and that has built-in pay-for-performance incentives, is an attractive social investment and a potential avenue for social mobility.

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